Pakka — AI PRD

An AI PRD for Pakka, an AI order copilot for small businesses that sell through WhatsApp

PRD Details

Product name	Pakka
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Status	Final for capstone submission

> **Pain Point:** Small businesses that take custom orders on WhatsApp negotiate and change each order across days of texts, images and payment screenshots, but the current agreed version of the order is never written down anywhere. Owners rely on memory, notebooks and scrolling, so details get missed as order volume grows or when more than one person replies, causing remakes, refunds and lost customers.

1. Executive Summary

What are we building?	A companion app that connects to a business's WhatsApp Business number and turns order conversations into live, trustworthy order tickets. Every detail links to its source message, changes are recorded in a history, and anything missing, conflicting or unverified is flagged. Pakka never replies to customers.

Why does it matter?	WhatsApp is the storefront for millions of small Indian businesses, and custom orders are its messiest workflow. One missed detail ruins an occasion and loses a customer. Meta's Coexistence feature, now supported in India, lets a business keep using the WhatsApp Business app while an approved partner reads the same chats through the API. Pakka can therefore work in the background without changing how owners sell.
Who is it for?	Primary: owner-operators of small custom-order businesses on WhatsApp Business, working alone or with one or two helpers. Secondary: helpers who reply from the same number. Beachhead segment: home bakers, followed by saree sellers, jewellery makers and cloud kitchens' direct WhatsApp orders.
How will we know it worked?	North Star: at least 98% of orders delivered with zero detail errors, owner-reported at delivery in the closed beta.
Prototype	**Deployed and working**: https://pakka-app.vercel.app/ . **Designs**: https://www.figma.com/design/3QEBLyRLp0qelizbC4MX8O/Capstone_BYOP_Pooja_Pakka?m=auto&t=kT7ShChLI1rQloyP-6
Code	https://github.com/poojamarivivekraja/pakka-app.git Seven scripted scenarios pass 150 of 150 checks across three runs.
Target launch	Capstone prototype: October 2026 (done). Closed beta with home bakers: Q1 2027, subject to Meta Tech Provider approval or a provider partnership.

2. Problem Statement and Background

The owner, or a helper replying from the same business number, manages each order by hand:

- Reads and replies to customer messages across days, including text, reference images, voice notes and payment screenshots.
- Copies the order into a notebook, spreadsheet or the phone's notes app at booking.
- Scrolls back through the chat to rebuild the latest version of the order before making it.
- Checks each payment screenshot against the bank or UPI app.

Evidence: conversations with 3 small business owners who sell through WhatsApp. They run the business on a dedicated WhatsApp Business number, often on a phone and a tablet. They move

Instagram enquiries to WhatsApp, and they track orders in a notebook, a workbook or the phone's notes app.

2.1 Current State

Trigger: a customer messages the business on WhatsApp, often after finding it on Instagram.

- **Step 1: Enquiry.** The customer shares the occasion, date and a rough idea; the owner asks follow-up questions.
- **Step 2: Requirements.** Size, flavour or material, design, personalisation text and references arrive in fragments, often with images.
- **Step 3: Price and confirmation.** The owner quotes, the customer negotiates, and they settle. The owner asks for an advance and the customer sends a screenshot.
- **Step 4: Changes.** Days later, the customer changes a detail. The owner or a helper agrees in chat. The notebook entry is often not updated.
- **Step 5: Fulfilment.** Before making and delivering the order, the owner scrolls back to reconstruct what was finally agreed, then coordinates time, address and the balance payment.

2.2 Pain Points

- **Details are scattered.** One order spans many messages, formats and days.
- **Changes silently override earlier details.** The latest agreed value isn't obvious, and copied notes go stale.
- **Knowledge is split across people.** A helper agrees to something the owner never sees.
- **Payment checks are manual and risky.** Screenshots must be matched against the bank app, and fake screenshots are a known scam.
- **Volume has a ceiling.** Beyond a handful of open orders, memory and notebooks stop working, so owners cap growth or accept errors.

2.3 Business Impact and Why Now

Cost of the status quo:

- **Direct cost:** a missed detail can mean remaking the product, refunding the customer, or both. For celebration orders the occasion can't be repeated, so the customer is often lost.
- **Reputation:** these businesses grow through referrals and Instagram, so one bad experience spreads.
- **Time:** owners spend time every day re-reading chats and re-copying notes, time taken from making products.
- **Growth ceiling:** owners can't take more orders without more mistakes or more staff.

Why now:

- **Coexistence.** Meta now lets a WhatsApp Business app number connect to the Cloud API without the owner changing how she works, and India is supported. Before this, reading app chats meant asking owners to abandon the app.
- **Multimodal AI.** Current models read mixed-language text, describe images and read payment screenshots reliably and cheaply enough for a small-business price point.
- **WhatsApp is already the storefront.** Owners and customers are there; Pakka adds no new channel.

2.4 Job To Be Done Framing

Job to be done	**When** I'm handling several custom orders over WhatsApp, each changing over days, **help me** always know the current, agreed version of every order and what needs my review, **so that I can** deliver every order exactly right without re-reading chats or keeping separate notes.
Trigger	A new order enquiry, any change to an open order, or preparing to make or deliver an order.
Outcome	The owner knows any order's current state in under a minute, with zero re-copying, and every delivered order matches what was agreed.
Why AI is justified	The input is unstructured, multimodal and mixed-language, and agreement is spread across a conversation, so rules and forms can't capture it, and customers won't fill in forms. Working out which order a message refers to, whether it changes something, and whether it was agreed needs language understanding. Deterministic code handles everything that needs no judgment: totals, required-field checks and status rules.

3. Opportunity and AI Capability Mapping

Opportunity	AI capability used
Decide whether a message is about an order, which order, and what kind (new detail, change, agreement, payment, small talk)	**Classification**

Pull order details into a structured ticket, each with the exact supporting words from the chat	**Information extraction (structured output)**
Detect changes, conflicts and agreement across the conversation	**Contextual reasoning over conversation history**
Describe reference images and read amounts and references from payment screenshots	**Vision (image understanding and OCR)**
Check required fields, compute balances, set statuses, verify quotes appear in source messages	**Deterministic rules (intentionally not AI)**

4. Personas and User Stories

4.1 User Personas

Persona	Description
Primary: Kavya, solo home baker	Runs a custom celebration cake business alone on a dedicated WhatsApp Business number; Instagram for discovery. **Goal:** deliver every order exactly as agreed and spend her time baking. **Frustration:** scrolling back to rebuild orders; changes that override earlier details; checking payment screenshots by hand.

Primary: Ayesha, home bakery owner with two helpers	Takes 20 to 30 custom orders a week; she and two helpers reply from one business number on a phone and a tablet. **Goal:** grow without losing control. **Frustration:** helpers agree to things she never sees; a shared notebook that's always out of date.
Secondary: Divya, part-time helper	Replies to customers while the owner is busy. **Goal:** answer confidently without contradicting earlier agreements. **Frustration:** no quick way to see an order's history before replying.

4.2 User Stories

As a...	**I want to...**	**So that...**	**Acceptance criteria**
Owner	see all open orders by due date, with what needs review first	I can plan my week at a glance	Given open orders exist, when I open Pakka, then I see them in week groups, each with a date block, a bold time, a status pill and a one-line summary of open issues, plus counts of open, needs-review and due-this-week orders
Owner	have details from the chat appear on the order automatically	I never re-copy anything	Given a customer states a detail and the business agrees, when the message syncs, then the field updates on the right order within 10 seconds, with a link to its source

Owner	see where any detail came from	I can trust the ticket	Given a field on a ticket, when I tap its source link, then I see the exact messages it came from, with the supporting words highlighted
Owner	know when something changed, without approving it again	I stay informed without double work	Given a change the business agreed in chat, when it syncs, then it applies automatically, shows "Updated from chat" with the previous value, and is logged in History, with no action required
Owner	see requests the business hasn't agreed to yet	I don't miss what a customer is waiting on	Given a customer requests a change and the business hasn't replied, when it syncs, then the current value stays and the ticket shows "Requested: [value], waiting for your reply"
Owner	be warned about payment screenshots that don't match	I don't lose money to errors or fakes	Given a screenshot amount differs from the advance requested, when it syncs, then the ticket shows "Payment screenshot doesn't match" and never shows the order as paid

Owner	confirm an order only when its required details are in	I never make something from an incomplete order	Given a required field is missing, when I view the ticket, then Confirm is disabled and the missing detail is named
Owner	be asked to choose when Pakka isn't sure	details never land on the wrong order or the wrong price	Given a customer has two open orders and the match is unclear, or two incompatible values appear, when the message syncs, then I see "Which order is this about?" or both values to choose from, and nothing is applied until I choose
Owner	correct any detail myself	the ticket is always right even when the AI is wrong	Given any field, when I edit it, then it saves and is marked "Edited by you" in the ticket and History
Helper	see an order's current state before replying	I don't contradict what was already agreed	Given I open a ticket, when it loads, then I see current values, recent updates and open issues, with no extra account needed

5. Model Strategy

Choosing the underlying model and build approach is an engineering decision, made jointly with the product team rather than by the product team alone.

What product owns:

- **Context requirements:** which inputs the model sees per message and which it never sees (section 7).
- **Approval gates:** the AI can update draft information, but only the owner can confirm an order, mark a payment as received, or resolve a conflict. The AI never messages customers.
- **Audit logging:** every model call is logged with the model, input and output tokens and latency. Every field keeps its source message, and every change is recorded in the order history.
- **Human-in-the-loop controls:** low-confidence extractions, unclear order matches and conflicts are routed to the owner and never resolved automatically. The owner can edit any field, and edits are marked "Edited by you".

Approach used in the prototype (agreed with engineering):

- **Model:** Gemini 3.5 Flash-Lite on the free tier with synthetic data only, one call per message and structured JSON output, with the model's thinking set to the lowest level it allows. In testing, the full Flash models were overloaded or very slow (80 to 110 seconds for a trivial prompt), while Flash-Lite answered in about 1.5 seconds and averages 1.6 seconds per call in the eval.
- **Adapter:** all model calls sit behind a provider-agnostic adapter, so changing the model is a configuration change. Rate-limit and overload errors are retried with backoff.
- **Production:** a paid API tier whose terms exclude training on customer data. Gemini Flash and Claude Sonnet will be compared on the eval set for accuracy, latency and cost.
- **Retrieval:** no fine-tuning and no RAG. The context per message (recent thread, open orders, business profile) is small enough to pass directly.
- **Configurability:** behaviour is controlled by a versioned system prompt plus the business profile, where each field has a plain-language description that tells the model what belongs in it. Sharpening these descriptions fixed the only run-to-run variance seen in testing.

6. Functional Requirements

ID	Requirement	Priority	Status
FR-1	Receive customer messages and business replies in real time, including text, images, edits, deletions and unreadable messages	Must	Done in the simulator; live webhook built, not yet tested with Meta

FR-2	Classify each message and link it to an existing order, a new order, or no order	Must	Done
FR-3	Extract the order fields defined by the business profile, each with the exact supporting quote; drop any extraction whose quote isn't found in the message	Must	Done
FR-4	Apply changes agreed in chat automatically, show "Updated from chat", keep the previous value, and log the change	Must	Done
FR-5	Hold customer requests the business hasn't agreed to as "Requested, waiting for your reply", without changing the current value	Must	Done
FR-6	Raise "Needs review" items for: missing required fields, conflicts, payment mismatches, low-confidence extractions, unclear order matches, and deleted or	Must	Done

	unreadable messages		
FR-7	Read payment screenshots (amount, reference, payee, date); never treat them as proof of payment; compute balances in code	Must	Done
FR-8	Attach reference images to the relevant order with a short description	Must	Done
FR-9	Show the exact source messages behind any field	Must	Done
FR-10	Allow confirmation only when all required fields are present; keep a confirmed order confirmed after an agreed change, marked "Updated after confirmation" until viewed	Must	Done
FR-11	Let the owner edit any field (marked "Edited by you"), mark payments as received, resolve issues, and mark orders delivered or cancelled	Must	Done

FR-12	Support business profiles that define fields, descriptions, field types and required flags; ship bakery (demo) and saree profiles	Should	Done
FR-13	Open the customer's WhatsApp chat from any order	Should	Done
FR-14	Notify the owner only when an order needs review, never for every message	Should	Mocked in the prototype (in-app toast)
FR-15	Show sync status, and a "Sync paused" banner when the WhatsApp connection drops	Should	Done (simulated in the prototype)
FR-16	Transcribe voice notes into the same pipeline	Could	Deferred
FR-17	Apply a customer's edit to the order the original message created; never create a second order from an edit	Must	Done
FR-18	Show a read-only list of customer chats, with a link to reply in WhatsApp	Should	Done

FR-19	Protect every action that changes data or calls the model: without the demo passcode the app is read-only, and the webhook accepts only requests carrying Meta's signature	Must	Done

7. Context and Data Pipeline

Context is the product. What goes in, and what stays out, defines what the AI can and cannot do.

7.1 What Goes In

Input	Source and notes
New message	WhatsApp webhook or simulator: text, image (with caption), edit or deletion event. Sender role marked "customer" or "business"
Recent conversation	The last 15 messages in this customer's chat, with message IDs and sender roles, so the model can judge agreement and changes
Customer's open orders	Current field values, IDs and open pending requests for that customer's open orders only
Business profile	Field keys, labels, types and plain-language descriptions for the business type
Time context	The message's date, weekday and timestamp, so relative dates ("this Saturday", "kal") can be resolved

7.2 Deliberately Excluded From Context

Every attribute excluded here is an attribute the AI cannot act on, intentionally or not. This is the first line of bias defence.

Excluded attribute	Reason for exclusion
Customer phone numbers and full contact names	Replaced with internal IDs; not needed to understand an order (data minimisation)
Other customers' chats and orders	Prevents cross-customer leakage and wrong-order links
Customer history beyond current open orders (past spend, frequency, location patterns)	The model treats every customer's order the same way; no profiling or prioritising by customer value
Group chats and personal chats	Not order channels; group chats aren't synced by WhatsApp's API
Bank or UPI account data	Pakka never accesses financial accounts; payment verification stays with the owner

7.3 Business Profiles

Pakka is generic across custom-order businesses through business profiles. Every profile shares universal fields and adds its own. Each field has a key, label, type, plain-language description and required flag.

Type	Fields
Universal	Occasion, items, due date, due time, fulfilment (pickup, delivery or courier), address, tracking number (courier), price agreed, advance requested (as stated by the business), balance (calculated by the app), reference images
Bakery (demo)	Size or weight, flavour, dietary, design notes, message on cake (exact text)

Saree	Fabric, colour, quantity, blouse stitching measurements, fall and pico, embroidery (exact text)
Jewellery (coming soon)	Metal, purity, weight, ring or bangle size, design reference, engraving (exact text)
Food orders on WhatsApp (coming soon)	Dishes and quantities, guests, spice level, veg or non-veg, packaging notes. Orders from Swiggy, Zomato and similar apps are excluded

Field types: exact text (spelling preserved, never corrected), measurement (always shown with its source), money (copied as stated, never calculated by the model), date, time, enum, items list.

8. Non-Functional Requirements

Category	**Requirement and target**
Performance	Ticket updated within 10 seconds of a message arriving (p95). The webhook is acknowledged within 2 seconds, with AI processing in the background. In the prototype eval, model calls averaged 1.6 seconds (p95 2.5 seconds)
Reliability / uptime	99.5% availability in beta. No message lost: deduplicated by WhatsApp message ID, failed processing retried with backoff, and failures shown to the owner rather than silently skipped
Security	Webhook signatures verified on the exact bytes received; with no app secret configured the webhook refuses everything. Access tokens stored server-side only. Encryption in transit and at rest. Owner login required in production. In the demo deployment the app is read-only without a passcode, and every action that changes data or calls the model needs it

Privacy and compliance	Designed for India's DPDP Act: the business is the data fiduciary and Pakka the data processor (to be confirmed by legal review). Explicit owner consent at onboarding, chat-history sharing opt-in, only order-relevant data stored, a defined retention period, deletion on disconnect. The prototype uses synthetic data only. Production must use a paid API tier whose terms exclude training on customer data. Message text and phone numbers are never written to logs
Scalability	One business handles 300+ messages a day without degraded latency. The architecture supports thousands of businesses on shared infrastructure
Usability	Mobile-first and installable as a web app. Body text at least 14 pt. Status never shown by colour alone. Tap targets at least 44 px. Works on phone, tablet and laptop
Integrations	WhatsApp Cloud API with Coexistence (production); simulator and a built, locally tested webhook (prototype); Gemini API; Supabase; Vercel

9. Behaviour, Output Specification and Workflow

This section specifies the behaviour, not just the output: what the AI does in every state, including when it is uncertain or wrong.

Key screens (designs: [Figma link]):

- **Orders:** week groups with date blocks and bold times, with "Needs review" shown first.
- **Order:** updates, issues, details with source links, payments and history.
- **Source sheet:** the exact messages behind any field.
- **Chats:** a read-only list of customers.
- **Your business:** the business profile and the details Pakka tracks.
- **Edge states:** sync paused, waiting for reply, conflict, which order, unreadable message.

9.1 Standard Output Per Order

Output field	Description
Order fields	Each profile field with its value, source message link, confidence, and the previous value if changed
Status	Needs review, Ready to confirm, Confirmed, Delivered, Cancelled

Needs review items	Issues requiring the owner, each with a plain-language explanation and a direct action
Updates from chat	Informational notes about agreed changes, dismissed with "Got it"
Payments	Price agreed, advance requested, screenshot amounts (unverified until the owner marks them received), and the balance calculated in code
History	A timestamped log of every change, its source, and who made it (chat or owner)

9.2 Required Behaviour States

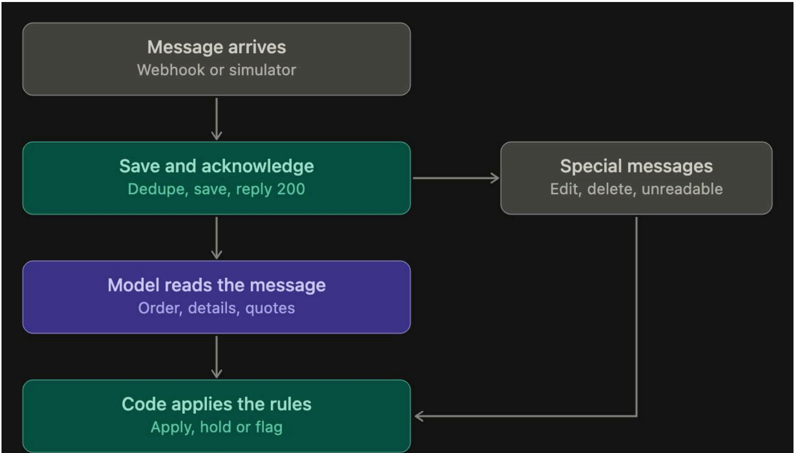
State	What triggers it	Required behaviour
Updated from chat	The customer requests a change and the business agrees in chat	Apply the change, show the previous value, log it, show an informational note. No action required
Requested, waiting for reply	The customer requests a change and the business hasn't replied	Keep the current value; show the requested value with "Reply in WhatsApp"
Low confidence	The model is unsure of a value, or a quote fails verification	Apply or hold as appropriate, and add "Check this" under Needs review, with the source
Conflict	Two incompatible values with no clear change (e.g. two prices)	Pick neither; show both under Needs review for the owner to choose

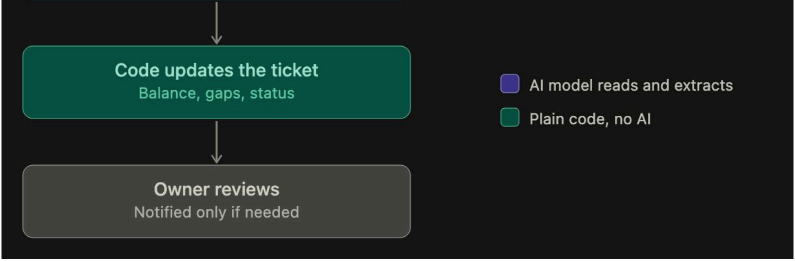
Unclear order match	The customer has multiple open orders and the message is ambiguous	Ask "Which order is this about?" and apply nothing until the owner chooses
Payment received (unverified)	A payment screenshot arrives	Record the amount and reference; mark it unverified; flag a mismatch if it differs from the advance requested; never show "paid"
Updated after confirmation	An agreed change arrives on a confirmed order	Keep it confirmed, apply the change, and mark "Updated after confirmation" until viewed
Edited message	The customer edits an earlier message	If the business hasn't replied since the original, apply the new values to the same order and add a "Customer edited a message" note. If it has replied, treat each changed value as a pending request. Never create a second order from an edit. Live delivery of edits depends on Meta: edits currently arrive as unsupported messages, so Pakka shows "1 message couldn't be read" until Meta restores the edit webhook. The edit logic is verified in the simulator
Deleted message	A deletion event	Flag the affected fields as needing a check
Unreadable message	Unsupported message event (error 131051 or 131060), including voice notes, stickers and locations	Flag "1 message couldn't be read; check WhatsApp"
Sync paused	WhatsApp disconnection event	Show a persistent banner on every screen until reconnected; never show stale data as current. Simulated in the prototype

9.3 End-to-End Decision Flow

- Step 1: A message arrives through the webhook (or the simulator); the signature is verified.
- Step 2: The message is deduplicated, saved, and acknowledged immediately.
- Step 3: Edits, deletions and unreadable messages branch to their own handling (9.2). An edit updates the order the original message created.
- Step 4: The system loads the customer's open orders, open pending requests, the last 15 messages and the business profile.
- Step 5: The model classifies the message and links it to an order, a new order, or none.
- Step 6: If the message isn't about an order, stop.
- Step 7: If the order match is unclear, raise "Which order is this about?" and wait for the owner.
- Step 8: The model extracts field updates with supporting quotes, plus any conflicts, agreements, image role and payment details.
- Step 9: Code verifies each quote appears in the message; failures become low-confidence items.
- Step 10: Code compares values with the current ticket. Changes agreed by the business apply as updates; unagreed requests are held as pending; conflicts go to Needs review.
- Step 11: For payment screenshots, code records the amount, recalculates the balance, and flags any mismatch.
- Step 12: Code rechecks required fields and recomputes the status.
- Step 13: The order history is written and the model call logged.
- Step 14: If anything now needs review, the owner is notified; the ticket updates on screen.

9.4 Flow Diagram





9.5 Copy Guidelines

- **Name agreements honestly.** Say "Agreed in chat by your business", not "You agreed", because WhatsApp doesn't identify which team member replied.
- **Never say "paid".** Use "Screenshot received · not verified" until the owner marks a payment as received.
- **Status labels:** Confirmed, Ready to confirm, Updated from chat, Needs review.
- **Describe issues plainly** and pair each with a direct action, for example "Address missing. Add address."
- **Label calculated numbers** with "Calculated by the app, not read from chat."

Failure mode	What goes wrong	Required guardrail	Owner
Hallucinated detail	The AI fills a field the chat never stated	Every extraction must include a quote verified in code; unsupported fields stay blank and are flagged	Engineering
Wrong order link	A detail lands on the customer's other order	Ambiguous matches go to the owner; links are shown on the ticket with sources	Product + Engineering

Misread change	An old value is kept or a new one misapplied	Changes show the previous value and source; unagreed requests are held as pending; low-confidence changes go to Needs review	Engineering
Edit creates a duplicate order	An edited message is read as a new order	Edits update the originating order whatever the model says; a duplicate-order check runs in every eval	Engineering
False payment confidence	A fake or wrong screenshot is treated as payment	Payments stay unverified until the owner marks them received; mismatches are flagged; "paid" is never shown	Product
Run-to-run variance	The model sometimes leaves a required field empty or reads two similar fields differently	A missing-field flag the owner can fill in; clear field descriptions in each profile; every scenario repeated three times in evals and reported as a pass rate	Engineering
Silent sync failure	The owner trusts a ticket that stopped updating	Sync status always visible; "Sync paused" banner; unreadable messages flagged	Engineering

Over-reliance	The owner stops checking and the AI is wrong	Confirm is a human action; required fields block it; every field links to its source; the in-app note says the ticket is built from synced messages	Product
Unauthorised changes or model use	Anyone with the public link edits data or spends the model quota	Read-only without the passcode; passcode on every write and model route; the webhook accepts only Meta-signed requests; automated checks fail if a route is left open	Engineering
Privacy breach	Customer data exposed or misused	Minimal context, no cross-customer data, encryption, retention limits, deletion on disconnect, no message text in logs, DPDP review	Product + Legal
AI acting on customers	The product sends messages or makes promises	Pakka has no capability to send customer messages	Product

11. Eval Criteria

Eval criteria define whether the AI is ready to launch. This is different from success metrics, which measure business outcomes after launch.

Prototype eval set: seven scripted scenarios (a clean order, an agreed change, an edited message, two orders from one customer, a price conflict, a payment mismatch, and small talk), with 50 automated checks. Each scenario is run three times and reported as a pass rate, because model output varies between runs. The scenarios were also used to find and fix problems, so the result shows the fixes hold, not how the model does on unseen messages.

Extended set (planned before beta): about 60 fresh messages, held out and never used for tuning, covering Hinglish, typos, relative dates, multiple open orders, deletions, images and other business profiles.

Eval dimension	Definition and threshold	Prototype result (14 Oct 2026)
Hallucinated fields	Fields with a value not supported by the chat: **0** in the eval set (launch blocker)	Not measured directly; every extraction is quote-checked in code
Required fields filled	Orders where every required field the chat states is filled: **at least 95%** of runs	All scenarios, 3 of 3 runs
Field extraction accuracy	Correct value in the correct field: **at least 95%**	All field checks passed in all runs (scripted scenarios only)
Exact text preservation	Personalisation text copied exactly, including spelling: **100%**	100% (S1 and S2, 3 of 3 runs)
Order linking accuracy	Messages linked to the correct order, with ambiguous cases routed to the owner rather than guessed: **at least 95%**	S4 (two orders, one customer): 3 of 3 runs

Change and agreement detection	Changes correctly classified as agreed, requested-only or conflict: **at least 95%**	S2 and S5: 3 of 3 runs
Edit handling	An edited message updates the existing order and never creates a duplicate: **100%**	S3: 3 of 3 runs, exactly one order
Payment reading	Amounts read correctly from screenshots: **at least 98%**, and **0** cases of an order shown as paid	S1 and S6: 3 of 3 runs; 0 orders shown as paid
Small talk false updates	Non-order messages that change a ticket: **2% or less**	S7: 3 of 3 runs, no changes
Language parity	Accuracy on Hinglish and mixed-language messages within **5 points** of English	Not yet measured
Latency	Ticket updated within **10 seconds** (p95)	Model calls: average 1.6 s, p95 2.5 s

Prototype results (gemini-3.5-flash-lite, 14 October 2026): seven scripted scenarios, 50 checks, each run three times; 150 of 150 checks passed. Model calls averaged 1.6 seconds (p95 2.5 seconds). Cost was about $0.0014 per message and $0.0076 per scripted order. Not yet measured: the extended held-out set, Hinglish parity, and results on a full Flash model. Full results: docs/EVAL_RESULTS.md.

12. North Star and Success Metrics

Layer	Metric	Target / threshold
North Star	Orders delivered with zero detail errors (owner-reported at delivery)	At least 98% in closed beta
Output (shallow)	Orders with a complete ticket at least 24 hours before due	At least 90%
Trust signal (AI-specific)	Fields kept without owner edits	At least 90%
Guardrail metric (AI-specific)	Owner-reported wrong details caused by Pakka	Fewer than 1 per 100 orders
User experience (AI-specific)	Time to know an order's current state; active days per week per owner	Under 1 minute; at least 4
Effort	Owners who stop keeping a separate notebook or notes app (survey)	At least 70% after 4 weeks
Cost (AI-specific)	AI and hosting cost per active business per month	About ₹300 or less at 100 orders a month. The current estimate is about ₹215 to ₹260 (measured token usage; about ₹96 per US dollar), rising to about ₹680 if usage triples

Pricing implication: at ₹499 a month the gross margin is about 49% at normal usage and negative if usage triples. A launch price of ₹799 gives about 68% at normal usage and stays positive under stress, with ₹499 kept as an entry plan that needs a fair-use limit on images. Willingness to pay is to be validated in beta.

13. Scope and MVP Boundary

13.1 Out of Scope (Permanently)

Out of scope item	Reason / when it may be revisited
Sending messages to customers	Pakka is a copilot, not an agent; the customer relationship stays with the owner. Not revisited
Verifying payments against bank accounts	Requires financial access and regulation; verification stays with the owner. Not revisited
Aggregator orders (Swiggy, Zomato and similar)	Those orders already arrive structured on the platforms; Pakka covers orders taken directly on WhatsApp. Not revisited
Group chats	Not synced by WhatsApp's API, and not an order channel for the target users

13.2 V1 Ships vs Later, Deferred

V1 ships	Later, explicitly deferred
Live order tickets from chats (simulator, plus a built webhook); source links and history; Needs review issues and "Updated from chat" notes; payment screenshot reading with mismatch flags; edit handling; chats view; owner editing, confirmation and	Voice note transcription; Instagram messages; individual helper accounts and roles; regional-language interface (for example Tamil and Hindi); real push notifications (an in-app toast in the prototype); jewellery and food-order profiles; live testing against Meta with Coexistence; the extended held-out eval set

delivery; bakery profile, plus a saree profile to show extensibility; passcode-protected demo	

14. Rollout and Release Strategy

An AI feature earns trust incrementally. Each stage below has a gate condition before moving forward.

Stage	Definition and gate condition
Stage 1: Alpha (capstone prototype)	Deployed prototype with a simulator, synthetic data and a built webhook tested against made-up payloads. **Gate:** all prototype eval checks pass (done: 150 of 150); live test against Meta and the extended held-out set still to do
Stage 2: Closed beta	10 to 20 home bakers through Tech Provider approval or a provider partnership, with weekly feedback calls. **Gate:** zero hallucinated-detail incidents reaching a delivered order; at least 90% of fields kept without edits; North Star at least 95%
Stage 3: Phased rollout by segment	Open to all home bakers, then saree and jewellery sellers, then cloud kitchens' direct WhatsApp orders, each with its own profile evals. **Gate:** each new profile meets section 11 thresholds before release
Stage 4: Full release	Public self-serve onboarding. **Gate:** North Star at least 98% across segments; support load sustainable

14.1 Kill Switch Conditions

Pause AI updates and fall back to read-only chat sync if any of the following occur:

- Any confirmed case of a hallucinated detail causing a wrong delivered order, until root-caused.
- Any case of an order shown as paid without the owner marking it received.
- Cross-customer data appearing on the wrong business's or customer's ticket.
- Wrong-order links above 3% in a rolling week.

15. Timeline and Milestones

Milestone	Target date	Owner / notes
Synopsis approved	Done	Pooja
Discovery worksheet and opportunity map	Done	Pooja
PRD, cost estimate and eval	Done, October 2026	Pooja
Prototype deployed	Done, October 2026	Pooja, using Claude Code
Presentation and demo video	October 2026, capstone deadline	Pooja
Live test against Meta; extended eval set	Before beta	Engineering
Closed beta	Q1 2027	Depends on the Tech Provider route (section 16)

16. Open Questions and Assumptions Log

Open question / assumption	Ticket type	Owner	How we resolve it
Become a Meta Tech Provider, or partner with an existing provider for Coexistence onboarding?	Spike	Product + Engineering	Compare timelines, cost and control; decide before beta

The webhook has been tested only with made-up payloads, not Meta's real servers	Test	Engineering	Test on a Cloud API number with the founder's own phone; adjust formats and the Graph API version
Meta currently delivers edited messages as unsupported, so live edits can't update orders	Dependency	Engineering	Track Meta's edit-webhook restoration; keep the unreadable-message flag as the fallback
How many owners reply from WhatsApp for Windows (an unsupported device)?	Research	Product	Ask beta owners; if common, add onboarding guidance to use supported devices
Assumption: owners will pay a small monthly fee. At ₹499 the margin is about 49% at normal usage and negative if usage triples	Research	Product	Willingness-to-pay interviews in beta; plan for ₹799 with a fair-use limit on images
Currency risk: model costs are in dollars, revenue in rupees	Research	Product	Review pricing as the exchange rate moves
Variance on lite models: required fields were sometimes left empty before the field descriptions were sharpened	Eval	Engineering	Keep repeating each scenario three times; run the held-out set; compare against a full Flash model when traffic is lower; switch only if the pass rate drops

The eval scenarios were used while fixing problems, so 150 of 150 doesn't measure unseen messages	Eval	Engineering	Build the held-out set of about 60 messages, never used for tuning
Accuracy on Tamil, Hindi and other regional-language messages	Eval	Engineering	Extend the eval set before expanding beyond Hinglish
DPDP roles and consent for customers' data	Legal review	Product + Legal	Confirm the processor role, consent wording and retention period before beta
Deleted or unreadable messages attach to the most recently updated open order, which may be the wrong one	Design	Product + Engineering	Attach them to the customer's chat and show them on that customer's open orders
Two messages from one customer arriving at the same moment are processed independently and could briefly conflict	Design	Engineering	Add a per-customer queue before beta
How do owners prefer to mark orders delivered?	Design	Product	Test a manual tap against detection from chat in beta